

Technical Data

Essential Specifications	LIBRA® 200FE	LIBRA® 200MC
Field Emitter	Schottky Field Emission System	Schottky Field Emission System with integrated OMEGA type monochromator
Accelerating voltage	200 kV	
Patented Koehler Illumination (parallel illumination)	Three-lens condenser system for parallel and homogeneous TEM wide-field illumination independent of illumination intensity Quick and easy switching between TEM and Spot mode by push-button operation Reproducible and controlled illumination conditions in TEM and Spot mode Condenser aperture automatically selected depending on magnification (AIS mode)	Three-lens condenser system for parallel and homogeneous TEM wide-field illumination independent of illumination intensity Quick and easy switching between TEM and Spot mode by push-button operation Reproducible and controlled illumination conditions in TEM and Spot mode
Spatial resolution Point resolution Information Limit STEM resolution	Truly symmetrical objective lenses = two variants: HR (High Resolution) HT (High Tilt) 0.24 nm 0.29 nm 0.14 nm 0.19 nm 0.30 nm 0.45 nm	
Energy resolution	≤ 0.7 eV	≤ 0.2 eV (Electrostatic OMEGA-shaped monochromator design, dispersion-free spot formation, 12µm/eV dispersion at filter slit plane, selectable slits with various widths)
Corrected OMEGA filter Alignment Dispersion Distortion Magnification factor Elements Detectors Acceptance angle Non-Isochromaticity	Corrected for 2nd order and optimized for 3rd order aberrations Factory alignment 1.85 µm/eV @ 200 kV < 1.5 % 1 (no magnification by filter) 4 magnets and 7 multi-pole correctors Selectable (4k x 4k SlowScan CCD camera possible) > 100 mrad @ ΔE = 10 eV slit width and 200 kV > 150 mrad @ ΔE = 20 eV slit width and 200 kV < 0.5 eV over Ø 2.5 µm field-of-view on sample	
Specimen stage Tilt range	5-axes fully eucentric goniometer α / β ± 30° / ± 30° (HR pole piece) α / β ± 70° / ± 30° (HT pole piece) both with double-tilt analytical holder α ± 70° (HR pole piece) with special tomography holder	
Imaging system	Two 3 lens projector groups in front and behind the energy filter	
Magnification	TEM 8 x - 1,000,000 x (for HR pole piece) STEM 2,000 x - 5,000,000 x EELS 20 x - 315 x (spectrum magnification)	
Vacuum system	Oil-free and differentially pumped vacuum system Pre-vacuum scroll pump with buffer tank TMP for viewing chamber, filter and airlock IGPs for the column 2 IGPs for the emitter area	
System control	WinTEM™ Graphical User Interface (GUI) based on Windows® XP, operated by mouse, keyboard and dedicated control panels	